Question ID: 343aaff6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

For the linear function f, the graph of $y = f(x)$ in the *xy*-plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f?

Answer

A. $f(x) = 5x$

B. $f(x) = 35x$

C. $f(x) = 7x + 5$

D. $f(x) = 12x + 5$

Correct Answer: C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the *xy*-plane and $(0, b)$ is the *y*-intercept of the graph. It's given that for the linear function f, the graph of $y = f(x)$ in the *xy*-plane has a slope of 7. Therefore, $m = 7$. It's also given that the graph of $y = f(x)$ in the *xy*-plane passes through the point $(0, 5)$. Therefore, the *y*-intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting 7 for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of 5, not 7, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice B is incorrect. This equation defines a function whose graph has a slope of 35, not 7, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a slope of 12, not 7.